

Characterize the exact worst-case approximation ratio for i.i.d. k -selection prophet inequalities

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Website #3 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Fix integers $n \geq k \geq 1$. We observe i.i.d. nonnegative random variables $X_1, \dots, X_n \sim F$ online, where F is known in advance. After seeing X_t , the decision-maker must irrevocably accept or reject it, and may accept at most k values in total. If a policy π accepts the random index set $S_\pi \subseteq [n]$, its reward is

$$\text{Val}(\pi, F) := \mathbb{E}_F \left[\sum_{t \in S_\pi} X_t \right].$$

The prophet benchmark is

$$\text{OPT}_{n,k}(F) := \mathbb{E}_F \left[\sum_{i=n-k+1}^n X_{(i)} \right],$$

the expected sum of the k largest order statistics.

The quantity of interest is

$$\gamma_{n,k} := \sup_{\pi} \inf_F \frac{\text{Val}(\pi, F)}{\text{OPT}_{n,k}(F)}.$$

The question asks for an analytical characterization of $\gamma_{n,k}$ for general $k \geq 2$, or at least of $\inf_{m \geq k} \gamma_{m,k}$, going beyond lower bounds extracted from the nonlinear differential-equation system of the recent literature.

What I prove instead is:

- a *general structural upper reduction*: the k -selection problem is pointwise upper-bounded by a neighboring *single-choice* problem whose prophet benchmark is the *average of the top k values*;
- an *exact variational characterization* of $\gamma_{n,n-1}$ (the one-rejection regime $k = n - 1$);
- explicit two-sided bounds in that regime, including a concrete bracket for the first nontrivial case $(n, k) = (3, 2)$.

Alternative readings considered

- **Does the policy know F ?** I adopt the standard prophet-inequality reading: yes, F is known and the policy may depend on it. Under any other reading, the quantifier structure changes substantially.
- **Do we allow arbitrary nonnegative laws?** I work with nonnegative i.i.d. laws F having $0 < \mathbb{E}_F[X] < \infty$. This is the natural regime in which the ratio is finite and nondegenerate. Continuity is not needed anywhere below; atoms are allowed.

Related literature found

- **Hill–Kertz (1982)** and **Correa–Fonca–Hoeksma–Oosterwijk–Vredeveld (2021)** — BACKGROUND. These are the canonical $k = 1$ references: Hill–Kertz derive the sharp i.i.d. constants implicitly, and Correa et al. match the tight bound algorithmically.
- **Brustle–Perez-Salazar–Verdugo (2025)** — PARTIAL. This is the closest source to the present problem: it introduces the nonlinear ODE system mentioned in the prompt and, importantly, states that an infinite-dimensional LP fully characterizes the fixed- (n, k) i.i.d. problem, but without a closed-form general solution.
- **Molina–Gast–Loiseau–Perchet (SODA 2025)** — PARTIAL. They solve a neighboring one-choice problem in which the prophet benchmark is the *average of the top ℓ* values, via an integral equation. This becomes directly relevant because of the reduction in part (i) of my theorem.
- **Kong–Murthy (2026 preprint)** — PARTIAL. This gives the strongest asymptotic lower bounds I found for the multiunit i.i.d. problem itself (distribution-sensitive, large n , large k); it still does not provide the exact worst-case-over- F formula asked here.
- **Livanos–Mehta (SODA 2024)** — BACKGROUND. This gives a precise characterization for the *cost-minimization* i.i.d. prophet problem. It looked promising for the $k = n - 1$ regime, but it does *not* directly apply because our objective is “total sum minus one rejected reward,” not “minimize the chosen cost.”

Proof sketch

There are two ideas.

1. A projection from k choices to one choice. Take any k -selection policy π . If one chooses uniformly at random an index $J \in \{1, \dots, k\}$, simulates π , and accepts the J -th item that π would accept (or nothing if π accepts fewer than J items), then the expected reward is exactly $1/k$ of π ’s reward, pathwise averaged over J . Therefore the pointwise optimal k -selection value can never exceed k times the pointwise optimal *single-choice* value against the benchmark $\text{OPT}_{n,k}(F)/k$, i.e. the average of the top k values. This gives a clean upper reduction from the original open problem to a solved neighboring one-choice benchmark.

2. Collapse of the state space when $k = n - 1$. If one may accept $n - 1$ out of n items, then one is equivalently choosing *exactly one item to reject*. For a fixed distribution F , this becomes a one-dimensional Bellman recursion: with m observations left, the optimal expected rejected value is r_m , where

$$r_1 = \mathbb{E}[X], \quad r_{m+1} = \mathbb{E}[\min(X, r_m)].$$

After normalizing $\mathbb{E}[X] = 1$, this recursion can be written purely in terms of the survival function $S(x) = \mathbb{P}(X > x)$:

$$r_{m+1} = \int_0^{r_m} S(x) dx.$$

The prophet loses only the minimum, whose expectation is $\int_0^\infty S(x)^n dx$. Thus $\gamma_{n,n-1}$ becomes an exact variational problem over mean-one survival functions.

To get an explicit lower bound, I compare the Bellman iterate r_n to $m_n := \int S^n$ by a repeated Hölder estimate. To get an explicit upper bound, I plug in the exponential distribution, for which the Bellman recursion closes exactly.

Main result

For a fixed law F , let $V_{n,k}^*(F)$ denote the optimal online value in the i.i.d. (k, n) -selection problem.

For the neighboring one-choice benchmark, define

$$B_{n,k}^*(F) := \sup_{\sigma \in \Pi_1(F)} \mathbb{E}_F[X_\sigma],$$

where $\Pi_1(F)$ is the class of one-selection online policies for the same horizon and same law F , and X_σ is the selected reward (taken as 0 if σ never selects).

Also let

$$\beta_{n,k} := \inf_F \frac{B_{n,k}^*(F)}{\text{OPT}_{n,k}(F)/k}.$$

For the one-rejection regime, let

$$\mathcal{S}_1 := \left\{ S : [0, \infty) \rightarrow [0, 1] : S \text{ is right-continuous, nonincreasing, and } \int_0^\infty S(x) dx = 1 \right\}.$$

For $S \in \mathcal{S}_1$, define

$$r_1(S) := 1, \quad r_{j+1}(S) := \int_0^{r_j(S)} S(x) dx \quad (j \geq 1),$$

and

$$m_n(S) := \int_0^\infty S(x)^n dx.$$

Theorem 1.

1. For every $n \geq k \geq 1$,

$$\gamma_{n,k} \leq \beta_{n,k}.$$

Equivalently, pointwise in F ,

$$V_{n,k}^*(F) \leq k B_{n,k}^*(F).$$

2. In the one-rejection regime $k = n - 1$,

$$\gamma_{n,n-1} = \inf_{S \in \mathcal{S}_1} \frac{n - r_n(S)}{n - m_n(S)}.$$

3. Define

$$\alpha_n := 1 - \left(1 - \frac{1}{n}\right)^{n-1}, \quad \eta_n := \min_{x \in [0,1]} \frac{n - x^{\alpha_n}}{n - x}.$$

Then

$$\gamma_{n,n-1} \geq \eta_n.$$

More explicitly,

$$\gamma_{n,n-1} \geq 1 - \frac{(1 - \alpha_n)\alpha_n^{\alpha_n/(1-\alpha_n)}}{n - 1}.$$

Hence, as $n \rightarrow \infty$,

$$\gamma_{n,n-1} \geq 1 - \frac{c_* + o(1)}{n - 1}, \quad c_* := e^{-1}(1 - e^{-1})^{e-1}.$$

4. Let $a_1 := 1$ and

$$a_{j+1} := 1 - e^{-a_j} \quad (j \geq 1).$$

Then

$$\gamma_{n,n-1} \leq \frac{n - a_n}{n - 1/n}.$$

In particular,

$$\min_{x \in [0,1]} \frac{3 - x^{5/9}}{3 - x} \leq \gamma_{3,2} \leq \frac{2 + e^{-1+e^{-1}}}{8/3} < 1.$$

Proof

Lemma 1 (Distribution-wise decomposition). *Let $\mathcal{F}$ be any class of input laws. Suppose that for each $F \in \mathcal{F}$ there is a nonempty set $\Pi(F)$ of admissible policies, and a payoff ratio $R(F, \pi)$ defined for $\pi \in \Pi(F)$. If a global policy means an F -indexed choice $(\pi_F)_{F \in \mathcal{F}}$ with $\pi_F \in \Pi(F)$, then*

$$\sup_{(\pi_F)_{F \in \mathcal{F}}} \inf_{F \in \mathcal{F}} R(F, \pi_F) = \inf_{F \in \mathcal{F}} \sup_{\pi \in \Pi(F)} R(F, \pi).$$

Proof. The inequality “ $\leq$ ” is immediate: for every global policy (π_F) ,

$$\inf_F R(F, \pi_F) \leq \inf_F \sup_{\pi \in \Pi(F)} R(F, \pi).$$

Taking the supremum over global policies gives

$$\sup_{(\pi_F)} \inf_F R(F, \pi_F) \leq \inf_F \sup_{\pi \in \Pi(F)} R(F, \pi).$$

For the reverse inequality, fix $\varepsilon > 0$. For each $F \in \mathcal{F}$, choose $\pi_F^\varepsilon \in \Pi(F)$ such that

$$R(F, \pi_F^\varepsilon) \geq \sup_{\pi \in \Pi(F)} R(F, \pi) - \varepsilon.$$

Then

$$\inf_F R(F, \pi_F^\varepsilon) \geq \inf_F \sup_{\pi \in \Pi(F)} R(F, \pi) - \varepsilon.$$

Taking the supremum over all global policies and then letting $\varepsilon \downarrow 0$ yields the reverse inequality. $\square$

Lemma 2 (Random acceptance-index projection). *Fix a law F and a k -selection policy π . There exists a one-selection policy σ on the same instance such that*

$$\mathbb{E}_F[X_\sigma] = \frac{1}{k} \mathbb{E}_F \left[\sum_{t \in S_\pi} X_t \right].$$

Proof. Sample J uniformly from $\{1, \dots, k\}$, independently of everything else. The policy σ simulates π online using the same internal randomness as π . It keeps track of how many items π would have accepted. When that count first reaches J , σ accepts that very item and stops. If π accepts fewer than J items overall, then σ accepts nothing.

This is a valid one-selection online policy: at every time, σ ’s decision depends only on J , the realized history, and the simulated state of π .

Now condition on a sample path of the values and on the internal randomness of π . Suppose π accepts exactly $m \leq k$ items, with accepted values in acceptance order $y_1, \dots, y_m$. Then conditional on that path,

$$\mathbb{E}[X_\sigma \mid y_1, \dots, y_m] = \frac{1}{k} \sum_{j=1}^m y_j,$$

because σ outputs y_j iff $J = j$, and outputs 0 when $J > m$. Taking expectation over the sample path and π 's randomness gives the claim. $\square$

Proof of part (1). Fix F . Let $V_{n,k}^*(F)$ be the optimal k -selection value and $B_{n,k}^*(F)$ the optimal one-selection value against the same horizon and same law. By the previous lemma, every k -selection policy π induces a one-selection policy σ with expected reward exactly $1/k$ of π 's reward. Therefore

$$B_{n,k}^*(F) \geq \frac{1}{k} V_{n,k}^*(F), \quad \text{so} \quad V_{n,k}^*(F) \leq k B_{n,k}^*(F).$$

Dividing by $\text{OPT}_{n,k}(F)$ yields

$$\frac{V_{n,k}^*(F)}{\text{OPT}_{n,k}(F)} \leq \frac{B_{n,k}^*(F)}{\text{OPT}_{n,k}(F)/k}.$$

Now apply the previous lemma with $R(F, \pi) = \text{Val}(\pi, F) / \text{OPT}_{n,k}(F)$ to identify

$$\gamma_{n,k} = \inf_F \frac{V_{n,k}^*(F)}{\text{OPT}_{n,k}(F)}.$$

Taking $\inf_F$ of the pointwise inequality gives

$$\gamma_{n,k} \leq \beta_{n,k}.$$

Notice that no i.i.d. assumption was used in the projection argument itself; only the definition of $\gamma_{n,k}$ uses the i.i.d. model. $\square$

Lemma 3 (Tail-integral identities). *Let $X \geq 0$ have survival function $S(x) = \mathbb{P}(X > x)$. Then for every $a \geq 0$,*

$$\mathbb{E}[\min(X, a)] = \int_0^a S(x) dx.$$

If $X_1, \dots, X_n$ are i.i.d. copies of X , then

$$\mathbb{E}[\min(X_1, \dots, X_n)] = \int_0^\infty S(x)^n dx.$$

Proof. For the first identity,

$$\mathbb{E}[\min(X, a)] = \int_0^\infty \mathbb{P}(\min(X, a) > x) dx = \int_0^a \mathbb{P}(X > x) dx = \int_0^a S(x) dx.$$

For the second identity,

$$\mathbb{E}[\min(X_1, \dots, X_n)] = \int_0^\infty \mathbb{P}(\min(X_1, \dots, X_n) > x) dx = \int_0^\infty S(x)^n dx,$$

because the X_i are independent. $\square$

Lemma 4 (Bellman recursion for one rejection). *Fix $n \geq 2$, and consider the regime $k = n - 1$. For a fixed law F with mean $\mu := \mathbb{E}_F[X] \in (0, \infty)$, let $r_m(F)$ denote the minimum possible expected value of the single rejected item when m i.i.d. observations from F remain and exactly one rejection must still be made. Then*

$$r_1(F) = \mu, \quad r_{m+1}(F) = \mathbb{E}_F[\min(X, r_m(F))] \quad (m \geq 1).$$

Consequently,

$$V_{n,n-1}^*(F) = n\mu - r_n(F).$$

Proof. If $m = 1$, there is only one remaining item and it must be rejected, so $r_1(F) = \mu$.

Now suppose $m \geq 1$ and we have $m + 1$ observations left. After observing the current value x , there are only two possibilities:

- reject x immediately, incurring rejected value x ;
- accept x and continue optimally on the remaining m observations, incurring expected rejected value $r_m(F)$.

Randomization cannot improve over taking the smaller of these two numbers, so the optimal conditional rejected value is $\min(x, r_m(F))$. Averaging over $x \sim F$ gives

$$r_{m+1}(F) = \mathbb{E}_F[\min(X, r_m(F))].$$

Finally, in the regime $k = n - 1$, every feasible policy accepts $n - 1$ items and rejects exactly one. Since the expected total sum of all n observations is $n\mu$, maximizing accepted reward is equivalent to minimizing expected rejected reward. Hence

$$V_{n,n-1}^*(F) = n\mu - r_n(F).$$

□

Proof of part (2). By the distribution-wise decomposition lemma,

$$\gamma_{n,n-1} = \inf_F \frac{V_{n,n-1}^*(F)}{\text{OPT}_{n,n-1}(F)}.$$

Because the ratio is homogeneous under positive scaling of the rewards, I may normalize $\mathbb{E}_F[X] = 1$. Let $S \in \mathcal{S}_1$ be the survival function of F . By the Bellman recursion and the tail-integral identity,

$$r_1(S) = 1, \quad r_{j+1}(S) = \int_0^{r_j(S)} S(x) dx.$$

Also,

$$\text{OPT}_{n,n-1}(F) = n - \mathbb{E}[\min(X_1, \dots, X_n)] = n - \int_0^\infty S(x)^n dx = n - m_n(S).$$

Therefore

$$\frac{V_{n,n-1}^*(F)}{\text{OPT}_{n,n-1}(F)} = \frac{n - r_n(S)}{n - m_n(S)}.$$

Taking the infimum over all mean-one survival functions $S \in \mathcal{S}_1$ gives

$$\gamma_{n,n-1} = \inf_{S \in \mathcal{S}_1} \frac{n - r_n(S)}{n - m_n(S)}.$$

□

Lemma 5 (Iterated Hölder bound). *Fix $n \geq 2$. Let $S \in \mathcal{S}_1$, and write*

$$M := m_n(S) = \int_0^\infty S(x)^n dx.$$

Then for every $a \geq 0$,

$$\int_0^a S(x) dx \leq a^{1-1/n} M^{1/n}.$$

Consequently,

$$r_n(S) \leq M^{\alpha_n}, \quad \alpha_n := 1 - \left(1 - \frac{1}{n}\right)^{n-1}.$$

Proof. Apply Hölder's inequality to $f = S\mathbf{1}_{[0,a]}$ and $g = \mathbf{1}_{[0,a]}$, with conjugate exponents n and $n/(n-1)$:

$$\int_0^a S(x) dx \leq \left(\int_0^a S(x)^n dx\right)^{1/n} \left(\int_0^a 1^{n/(n-1)} dx\right)^{(n-1)/n} \leq M^{1/n} a^{1-1/n}.$$

Now define exponents b_j by

$$b_1 := 0, \quad b_{j+1} := \frac{1}{n} + \left(1 - \frac{1}{n}\right)b_j.$$

I claim that

$$r_j(S) \leq M^{b_j} \quad (j \geq 1).$$

This is true for $j = 1$ because $r_1(S) = 1 = M^0$. If it holds for j , then

$$r_{j+1}(S) = \int_0^{r_j(S)} S(x) dx \leq r_j(S)^{1-1/n} M^{1/n} \leq M^{(1-1/n)b_j + 1/n} = M^{b_{j+1}}.$$

Thus the claim follows by induction.

It remains to solve the linear recursion for b_j . A direct check gives

$$b_j = 1 - \left(1 - \frac{1}{n}\right)^{j-1}.$$

Hence

$$r_n(S) \leq M^{1-(1-1/n)^{n-1}} = M^{\alpha_n}.$$

□

Proof of part (3). Fix $S \in \mathcal{S}_1$, and write $M = m_n(S) \in [0, 1]$. By the previous lemma,

$$r_n(S) \leq M^{\alpha_n}.$$

Therefore the pointwise optimal ratio from part (2) satisfies

$$\frac{n - r_n(S)}{n - m_n(S)} \geq \frac{n - M^{\alpha_n}}{n - M}.$$

Taking the infimum over S yields

$$\gamma_{n,n-1} \geq \min_{x \in [0,1]} \frac{n - x^{\alpha_n}}{n - x} = \eta_n.$$

For the explicit bound, write

$$1 - \frac{n - x^{\alpha_n}}{n - x} = \frac{x^{\alpha_n} - x}{n - x} \leq \frac{x^{\alpha_n} - x}{n - 1} \quad (0 \leq x \leq 1).$$

For fixed $\alpha \in (0, 1)$, the function $h_\alpha(x) := x^\alpha - x$ on $[0, 1]$ attains its maximum where

$$h'_\alpha(x) = \alpha x^{\alpha-1} - 1 = 0,$$

namely at

$$x_* = \alpha^{1/(1-\alpha)}.$$

Its maximum value is

$$h_\alpha(x_*) = x_*^\alpha - x_* = (1 - \alpha)\alpha^{\alpha/(1-\alpha)}.$$

Therefore

$$\eta_n \geq 1 - \frac{(1 - \alpha_n)\alpha_n^{\alpha_n/(1-\alpha_n)}}{n - 1}.$$

Since $\alpha_n \rightarrow 1 - e^{-1}$ as $n \rightarrow \infty$, this becomes

$$\gamma_{n,n-1} \geq 1 - \frac{c_* + o(1)}{n - 1}, \quad c_* := e^{-1}(1 - e^{-1})e^{-1}.$$

□

Proof of part (4). Take F to be the exponential distribution with mean 1, so $S(x) = e^{-x}$. By part (2), the optimal ratio for this fixed F is

$$\frac{n - r_n(S)}{n - m_n(S)}.$$

Here

$$m_n(S) = \int_0^\infty e^{-nx} dx = \frac{1}{n}.$$

Also, the Bellman recursion becomes

$$r_{j+1}(S) = \int_0^{r_j(S)} e^{-x} dx = 1 - e^{-r_j(S)}.$$

Thus if we write $a_1 := 1$ and $a_{j+1} := 1 - e^{-a_j}$, then $r_j(S) = a_j$ for all j , and so the optimal ratio for the exponential law is

$$\frac{n - a_n}{n - 1/n}.$$

Since $\gamma_{n,n-1}$ is the infimum over all laws, we obtain

$$\gamma_{n,n-1} \leq \frac{n - a_n}{n - 1/n}.$$

For $n = 3$,

$$a_2 = 1 - e^{-1}, \quad a_3 = 1 - e^{-(1-e^{-1})} = 1 - e^{-1+e^{-1}},$$

hence

$$\gamma_{3,2} \leq \frac{3 - a_3}{3 - 1/3} = \frac{2 + e^{-1+e^{-1}}}{8/3}.$$

Combining this with part (3), where $\alpha_3 = 1 - (2/3)^2 = 5/9$, gives

$$\min_{x \in [0,1]} \frac{3 - x^{5/9}}{3 - x} \leq \gamma_{3,2} \leq \frac{2 + e^{-1+e^{-1}}}{8/3} < 1.$$

□

Gap to the full statement

The full open problem remains unsolved. Concretely, what is still missing is:

- an exact general formula for $\gamma_{n,k}$ when $2 \leq k \leq n-2$;
- a proof that the upper reduction in part (1) is tight, or a counterexample showing a strict gap;
- an optimization of the exact variational formula in part (2), which I only accomplished up to explicit bounds, not an exact minimizing law;
- a matching asymptotic characterization of $\inf_{m \geq k} \gamma_{m,k}$ as a function of k .

Still, part (1) gives a clean route to an *analytic upper envelope*: once one imports the integral-equation characterization of the neighboring one-choice top- k -average benchmark from Molina et al., one immediately gets an analytically characterized upper bound on $\inf_{m \geq k} \gamma_{m,k}$. Meanwhile, parts (2)–(4) solve the $n-k=1$ regime down to a one-dimensional variational problem and explicit two-sided estimates.

Self red-team

- **Counterexample attempt against part (1).** I tested the pathwise identity in the tricky case where π accepts only $m < k$ items. The projected one-choice policy then outputs 0 with probability $(k-m)/k$, so its conditional expected reward is exactly $(1/k) \sum_{j=1}^m y_j$, not $(1/m) \sum y_j$. This is the correct identity, and the proof survives.
- **Counterexample attempt against the Bellman recursion.** For a Bernoulli law $X \in \{0,1\}$ with $\mathbb{P}(X=1)=p$, the recursion gives $r_1=p$, $r_{m+1}=pr_m$, hence $r_n=p^n$. The prophet’s minimum also has expectation p^n , so the ratio is 1. This matches the intuitive optimal policy “reject the first 0 if one appears; otherwise reject the last 1.” So the recursion behaves correctly on a discrete law with atoms.
- **Counterexample attempt against the tail-integral formulas.** I checked both a continuous law (exponential) and an atomic law (Bernoulli). The formulas

$$\mathbb{E}[\min(X, a)] = \int_0^a S(x) dx, \quad \mathbb{E}[\min(X_1, \dots, X_n)] = \int_0^\infty S(x)^n dx$$

hold in both cases; continuity is not needed.

- **Three plausible referee objections.**
 1. “*You swapped sup and inf illegitimately.*” I addressed this by isolating the distribution-wise decomposition lemma. The swap is valid only because the policy is allowed to depend on the known distribution F ; I stated this reading explicitly.
 2. “*The projected one-choice policy may need recall.*” It does not. It stops exactly when the simulated k -choice policy reaches its J -th acceptance. No accepted item is revisited.
 3. “*Your Hölder step may hide extra regularity assumptions.*” It does not. The only facts used are $S \in [0,1]$, S nonincreasing, and $\int_0^\infty S = 1$. In particular $S \in L^n$ because $\int S^n \leq \int S = 1$.

- **Boundary checks.**

- If $n = k$, then $\gamma_{n,n} = 1$, as it should be.
- In the special regime $k = n - 1$, the exact formula in part (2) applies even for $n = 2$, but the lower bound in part (3) is *not* tight there; I do not claim otherwise.
- The exponential upper bound in part (4) shows $\gamma_{3,2} < 1$, so the near-full-capacity regime does *not* trivialize to perfect prophet performance.

Ideas tried

1. **Try to prove equality in part (1).** Idea: perhaps every one-choice policy against the average-top- k benchmark can be lifted to a k -choice policy with the same ratio, which would solve the original problem by reduction to Molina et al.’s one-choice model. Obstruction: I could not find a converse projection. The top k order statistics can cluster in time, and naive repetition of a one-choice policy does not preserve the same benchmark comparison.
2. **Optimize the exact variational formula for $k = n - 1$ by calculus of variations.** Idea: use the explicit functional

$$S \mapsto \frac{n - r_n(S)}{n - m_n(S)}$$

over $\mathcal{S}_1$, derive Euler–Lagrange conditions, and identify an extremal survival function. Obstruction: the nested map $r_n(S)$ is highly nonlocal; even for $n = 3$, the stationarity condition becomes awkward, and I did not isolate a manageable extremizer class.

3. **Generalize the one-rejection collapse to two rejections ($n - k = 2$).** Idea: encode the problem as minimizing the expected sum of two rejected items, hoping for a two-dimensional recursion that might still be analytically tractable. Obstruction: the Bellman state becomes genuinely two-dimensional, and the monotonicity/Hölder argument that worked in one dimension no longer closed cleanly.
4. **Construct a sharp hard family for general k using rare very-high / medium / low values.** Idea: derive explicit upper bounds by solving the dynamic program on a three-point law and then optimizing the parameters. Obstruction: the dynamic program for general k on such laws is analyzable numerically but I did not obtain a closed symbolic optimization, so I did not include it as a theorem.

Invoked results

- **Hölder’s inequality** (T00L, classical, used once, not imported from a paper). Full statement used: if $p, q > 1$ are conjugate exponents and $f \in L^p, g \in L^q$, then

$$\int |fg| \leq \|f\|_{L^p} \|g\|_{L^q}.$$

Hypothesis audit in the present setting: $p = n, q = n/(n - 1), f = S\mathbf{1}_{[0,a]}, g = \mathbf{1}_{[0,a]}$; since $0 \leq S \leq 1$ and $\int_0^\infty S = 1$, we have $S \in L^n$, and $\mathbf{1}_{[0,a]} \in L^{n/(n-1)}$. So the inequality applies exactly as written.

- No external theorem is used as a black box inside the main proof. All Bellman recursions and tail-integral identities are derived in-line.

Self-confidence

3

I checked every mathematical step in the stated theorems and deliberately stress-tested the quantifier swap, the projection argument, and the Bellman recursion; the only reason this is not a 4 is that the full problem remains open and my special-case bounds are not optimized to tightness.